

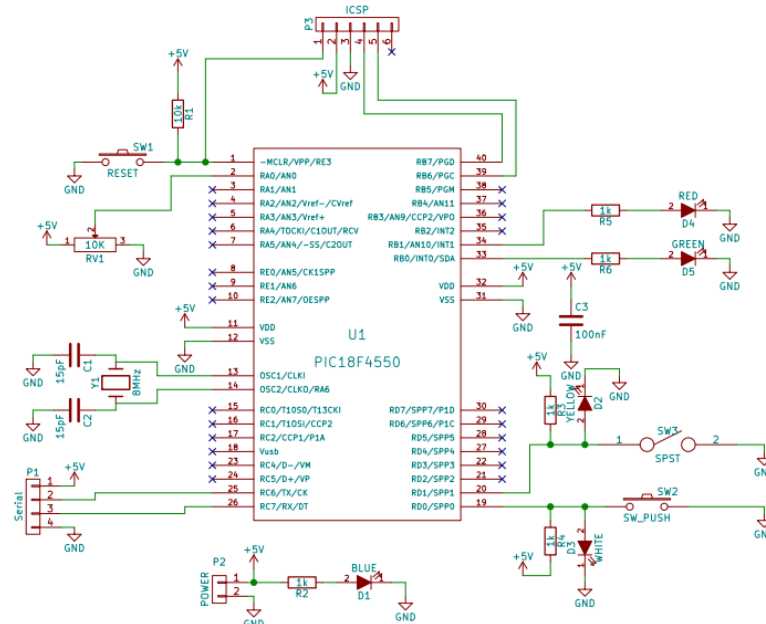
Course reference books:

- **Theory reference book:** microelectronics-circuit-analysis and design by Neamen
- **Eagle for PCB design book:** Designing Circuit Boards with EAGLE

Mini Project: Block diagram/Flowchart, Schematic, Simulation, PCB design, Hardware.

- Ashby _Circuit_Design. _Know_It_All
- **Programmable IC Simulator Laboratory (PICSimLab).**
- **Fritzing** is an electronics design and prototyping platform for electronic projects and educators.

Homework_3 next week: Design a PCB circuit for an 8/16 or 32-bit microcontroller containing more than four subcircuits. Some examples of these subcircuits are the oscillator, voltage regulator, and reset switch, as well as push-bottom switches (KEY) and a seven-segment display unit using a bus bar connection or LEDs for display signal.



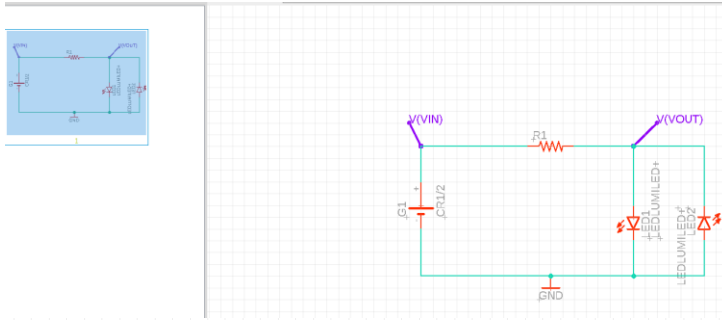
DRC and ERC:

Electrical rule check (ERC): Tools capable of identify any defects in the circuit that may cause malfunctions after delivery. They also ensure that integrated circuits function properly and that PCBs are manufactured with the necessary reliability and capacity. Also ensures that the design follows a set of electrical rules specified by the industry standards.

Design rule check (DRC): Ensuring the integrated circuit's layout complies with the design specifications and guidelines provided by the design team. These guidelines specify layout restrictions like wire width, component spacing, and minimum and maximum dimensions. Check for consistency between the integrated circuit's logical and physical layouts. This guarantees that the circuit connectivity that the engineers planned is reflected appropriately in the design.

Procedure to convert non-spice schematic to a spice-schematic

Create a series circuit from independent power supply with resistor and two LED diodes.



Add Model

Verify the spice types for all parts below, and map spice models to those parts not already mapped. The DONE column will show a green checkmark when all parts are properly updated. Click Done when you have mapped all parts in the table.

Done	Part	Gate	Spice Type	Value	Unit	Model	Map
!	G1	1	J: Junction Field-Effect Transistor	CR1/2			MAP
✓	R1	G\$1	R: Resistor		ohm		MAP
!	LED1	G\$1	L: Inductor	LEDLUMIL...	henry [H]		MAP
✓	GND1	1	Ground / Power Pin	GND			MAP
!	LED2	G\$1	L: Inductor	LEDLUMIL...	henry [H]		MAP

Done

Add Model

Verify the spice types for all parts below, and map spice models to those parts not already mapped. The DONE column will show a green checkmark when all parts are properly updated. Click Done when you have mapped all parts in the table.

Done	Part	Gate	Spice Type	Value	Unit	Model	Map
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✓	R1	G\$1	R: Resistor		ohm		MAP
!	LED1	G\$1	D: Diode	LEDLUMIL...			MAP
✓	GND1	1	Ground / Power Pin	GND			MAP
!	LED2	G\$1	L: Inductor	LEDLUMIL...	henry [H]		MAP

Done

Add Model

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✓	R1	G\$1	R: Resistor		ohm		MAP
!	LED1	G\$1	D: Diode	LEDLUMIL...			MAP
✓	GND1	1	Ground / Power Pin	GND			MAP
!	LED2	G\$1	D: Diode	LEDLUMIL...			MAP

Done

Map To Model

Load Model | Map Pins

Model type: [Intrinsic Device / No Model]

Map

OK

Add Model

Verify the spice types for all parts below, and map spice models to those parts not already mapped. The DONE column will show a green checkmark when all parts are properly updated. Click Done when you have mapped all parts in the table.

Done	Part	Gate	Spice Type	Value	Unit	Model	Map
!	G1	G\$1	M: MOSFET	3v			MAP
✓	R1	G\$1	R: Resistor		ohm		MAP
✓	LED1	G\$1	D: Diode	LEDLUMIL...		DMOD	MAP
✓	GND1	1	Ground / Power Pin	GND			MAP
!	LED2	G\$1	D: Diode	LEDLUMIL...			MAP

Done

